

TimbreEdit-Bench

Reference-conditioned editing with paired structured and descriptive instructions

01 Controlled audio pairs

200 profiles · 9 timbre axes

Attribute	Source	Target
Pitch (Low/Mid/High)	mid	low
Brightness (Dark/Bright/...)	dark	bright
Other 7 axes (Gender/Age/...)	unchanged	

Profile differences specify the edits

7 edit types · 22 sampling strata

Voice-design TTS

Source | Target

Chinese/English · 24 kHz

Instruction adherence

Requested attributes reach their target

02 Dual-level instructions

L1 · Structured Instructions

- Pitch register: mid → low.
- Brightness: dark → bright.
- Others: same.

L2 · Descriptive Instructions

Transform this dark, mid-register voice into a distinctly brighter voice with a lower pitch.

(Prompt) Rewrite the structured instruction as a natural voice-editing request, preserving all changes without adding new ones.

Attribute preservation

Unedited attributes remain unchanged

2200 candidate samples · Chinese/English

03 Quality screening

Instruction Checks

Human Annotators

Programmatic checks

Cross-model review

Rule-based decision

Curated Benchmark

Rejected L2: revise and retry (up to 5 rounds)

Audio Pair + L1 Edit

- Correct changes?
- Other traits preserved?

Pass?

Yes

✓ Keep

No

✗ Rejected

L1 vs. L2 robustness

Compare structured and natural instructions

IntraTTS-Bench

Within-utterance local instruction control

Multi-instruction Control

LLM text + segment instructions

TTS Synthesis

One continuous utterance · 2-3 controlled segments

One continuous utterance · 2-3 controlled segments

Human

- Pass
- Fail

Gemini

- Pass
- Fail

Segment Control + Voice Consistency